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molecular differences, mainly in DNA sequences, to gain information and to establish genetic relationship between the members of different taxonomic categories. The advent of DNA cloning and sequencing methods have contributed immensely to the development of molecular taxonomy and population genetics over the years. These modern methods have revolutionised the field of molecular taxonomy and population genetics with improved analytical power and precision.

The results of a molecular phylogenetic analysis are expressed in

Molecular Markers

Allozyme electrophoresis is a method which can identify genetic variation at the level of enzymes that are directly encoded by DNA.

Mitochondrial DNA markers are non- nuclear DNA located within organelles in the cytoplasm called mitochondria. The entire genome undergoes transcription as one single unit. They are not subjected to recombination and thus they are homologous markers.

Microsatellite is a simple DNA sequence which is repeated several times across various points in the DNA of an organism. These (usually 2-5) repeats are highly variable and these loci can be used as markers. (Example: TGTGTG, in which two base pairs repeat, the region are termed **tandem repeat**.)

Single nucleotide polymorphisms arise due to single nucleotide substitutions (transitions/transversions) or single nucleotide insertions/deletions. SNPs are the most abundant polymorphisms in the genome (coding and non-coding) of any organism. These single nucleotide variants can be detected using PCR, microchip arrays or fluorescence technology.

DNA microarray or DNA chip consists of small glass microscope slides, silicon chip or nylon membranes with many immobilized DNA fragments arranged in a standard pattern. A DNA microarray can be utilized as a medium for matching a reporter probe of known sequence against the DNA isolated from the target sample which is of unknown origin. Species-specific DNA sequences could be incorporated to a DNA microarray and this could be used for identification purposes.

Arbitrary markers are sometimes used to target a segment of DNA of unknown function. The widely used methods of amplifying unknown regions are RAPD and AFLP DNA.

Specific Nuclear DNA markers: Variable Number of Tandem Repeat is a segment of DNA that is repeated tens or even hundreds to thousands of times in nuclear genome. They repeat in tandem; vary in number in different loci and differently in individuals. There are two main classes of repetitive and highly polymorphic DNA viz. **minisatellite** DNA referring to genetic loci with repeats of length 9-65 bp and microsatellite DNA with repeats of 2-8 bp (1-6) long. Microsatellites are much more numerous in the genome of vertebrates than minisatellites.